Discrete Response, Time Series and Panel Data - Exercises (Rep.)

Nils Rechberger

2026-03-05

Series 1

Exercise 1.1

a

12

b

None

c

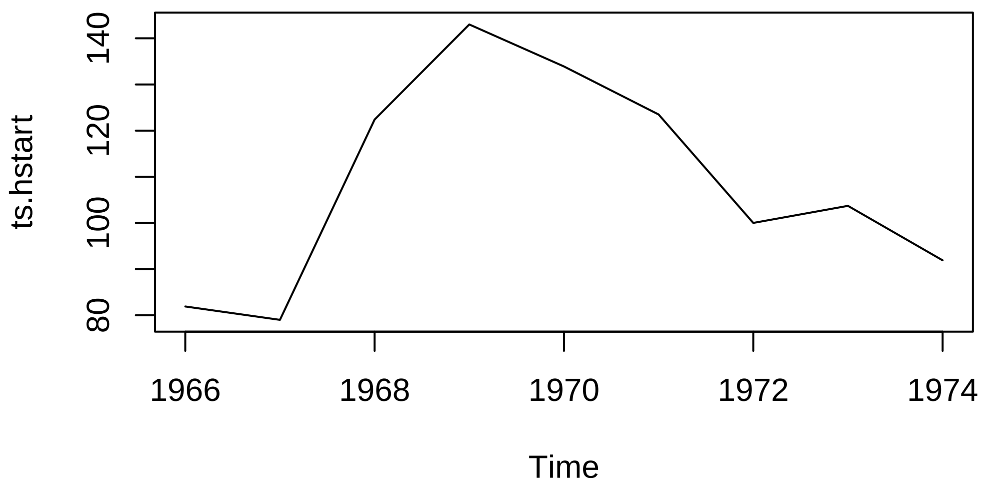
7

d

365

Exercise 1.2

```r
hstart <- read.table("/home/nils/dev/mscids-notes/fs26/rtp/data/hstart.dat")

ts.hstart <- ts(
  data=hstart,
  start=1966,
  end=1974
  )

plot(ts.hstart)
```



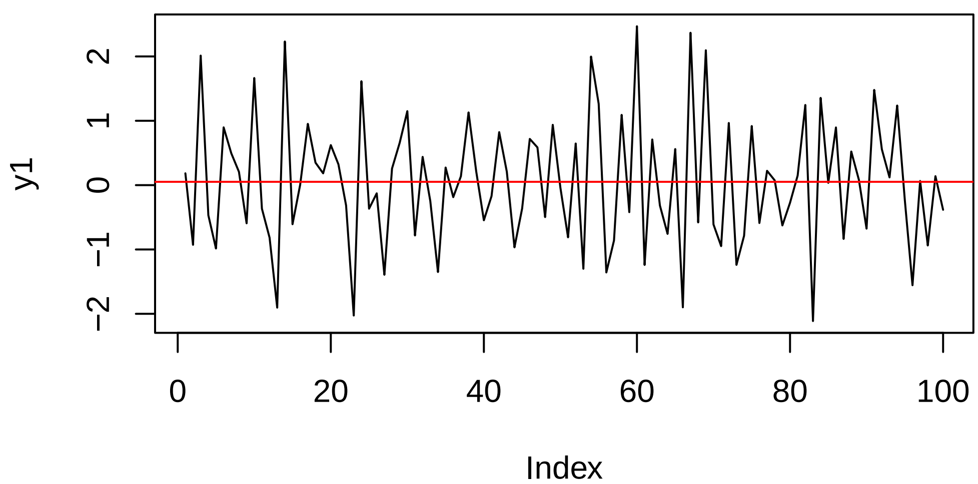
The process is not stationary. We see a seasonal trend from 1967 to 1972.

Exxercise 1.3

a

```r
set.seed(1)

Et <- ts(rnorm(101, 0, 1))
Et[1] <- 0
y1 <- 0 #delete later
```

```
for (i in 2:length(Et)) {
y1[i] <- Et[i] - 0.5 * Et[i-1]
}

y1 = y1[2:length(y1)]
ts.y1 = ts(y1)
plot(y1, type="l")
abline(h=mean(y1), col="red")
```

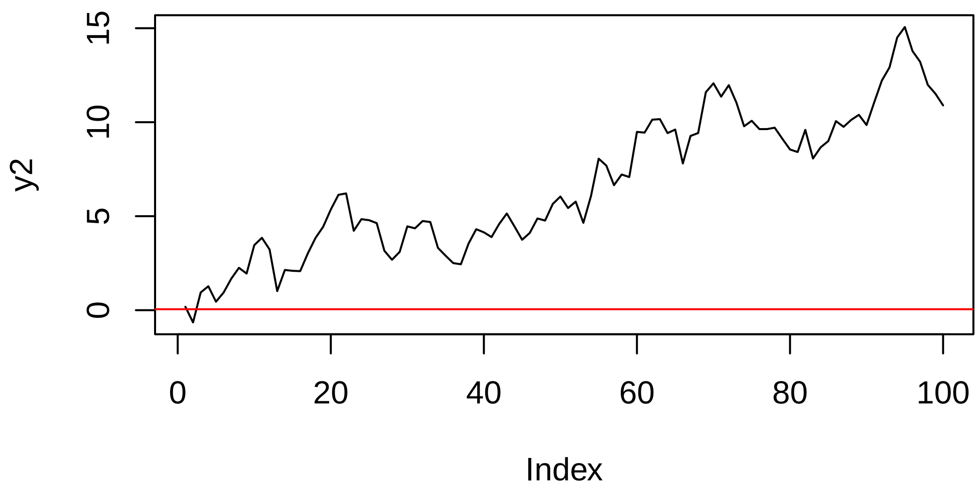


The process seems stationary.

b

```
set.seed(1)

Et <- ts(rnorm(101, 0, 1))
Et[1] <- 0
y2 <- 0 #delete later

for (i in 2:length(Et)) {
```

```
y2[i] <- y2[i-1] + Et[i]
}

y2 = y2[2:length(y2)]
ts.y2 = ts(y2)
plot(y2, type="l")
abline(h=mean(y1), col="red")
```

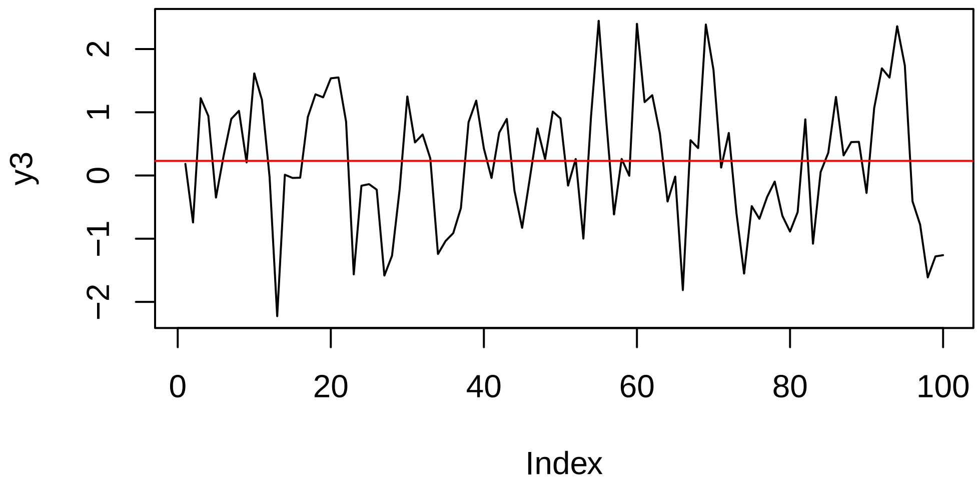


The process is not stationary.

c

```
set.seed(1)

Et <- ts(rnorm(101, 0, 1))
Et[1] <- 0
y3 <- 0 #delete later

for (i in 2:length(Et)) {
y3[i] <- 0.5 * y3[i-1] + Et[i]
}
```

```r
y3 = y3[2:length(y3)]
ts.y3 = ts(y3)
plot(y3, type="l")
abline(h=mean(y3), col="red")
```

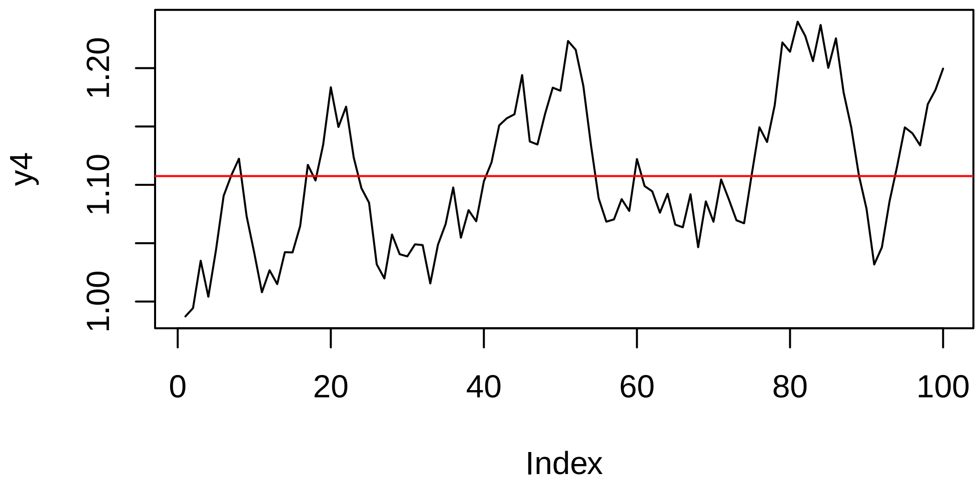


The process seems stationary.

d

```r
set.seed(1)

Et <- ts(runif(101, 0.95, 1.05))
Et[1] <- 0
y4 <- 1 #delete later

for (i in 2:length(Et)) {
y4[i] <- y4[i-1] * Et[i]
}

y4 = y4[2:length(y4)]
```

```
ts.y4 = ts(y4)
plot(y4, type="l")
abline(h=mean(y4), col="red")
```



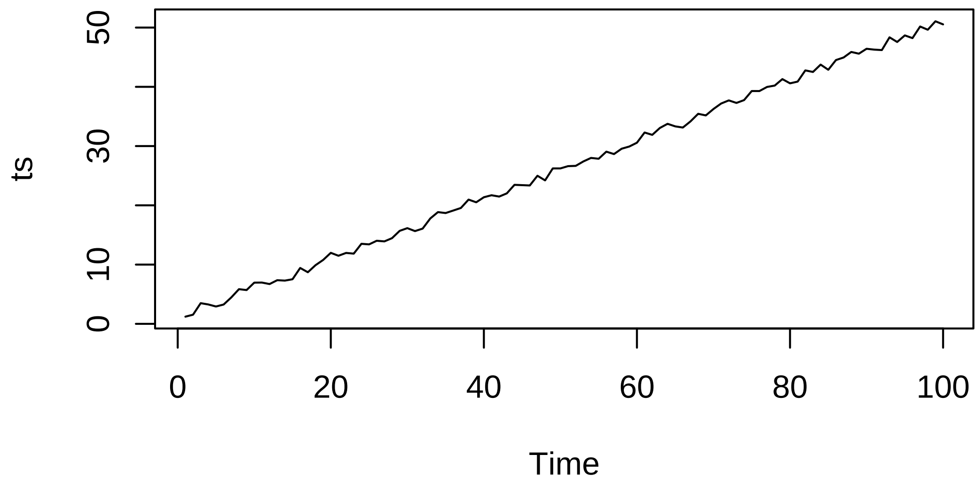
The model is not stationary

Series 2

Exercise 2.1

a

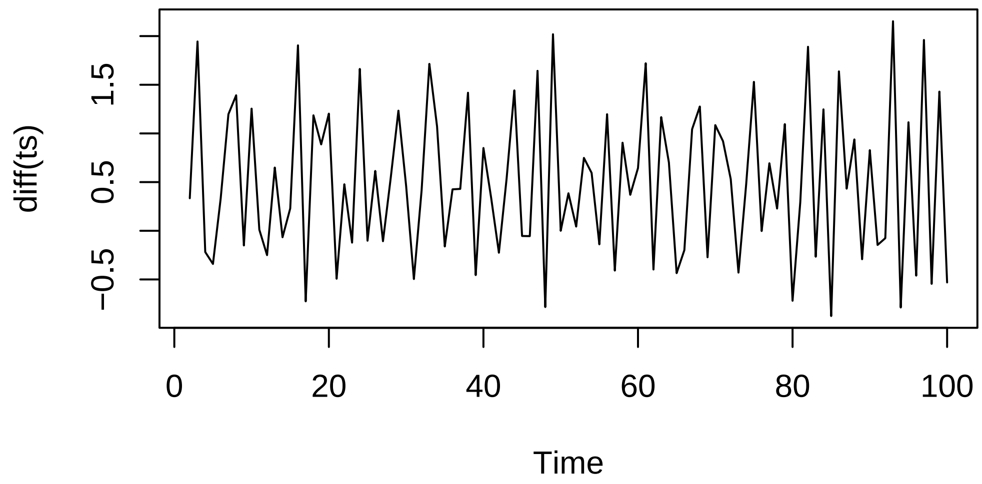
```
t <- seq(1, 100, length = 100)
data <- 0.5 * t + 1 + runif(100, -1, 1)
ts <- ts(data)

plot(ts)
```




```r
print(length(ts))
```

```
[1] 100
```

```r
plot(diff(ts))
```



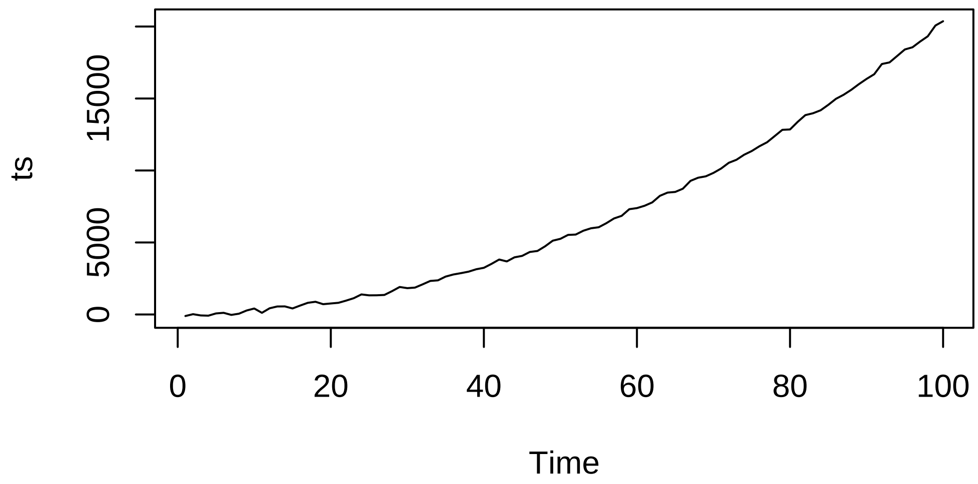

```
print(length(diff(ts)))
```

```
[1] 99
```

We can remove the linear trend with one time backward differencing. The length of the ts increases by 1.

b

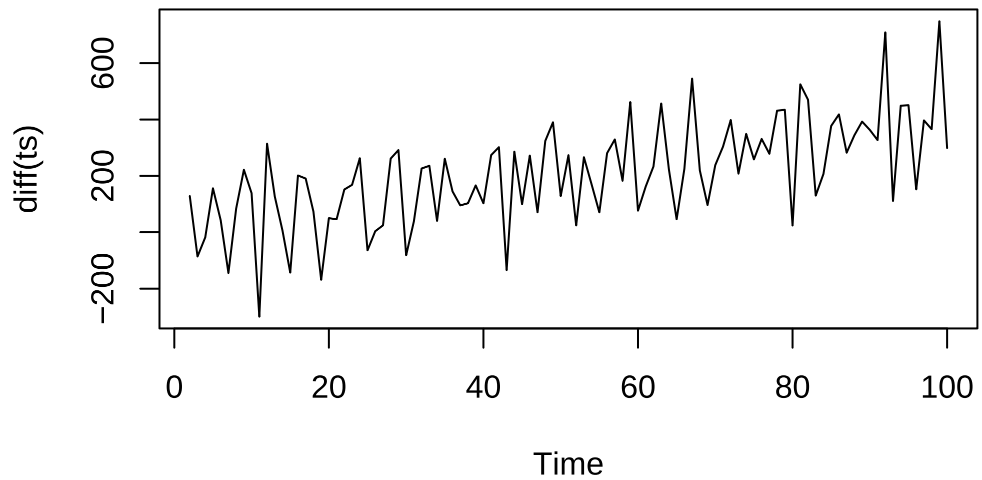
```
t <- seq(1, 100, length = 100)
data <- 2 * t**2 + 3 * t - 1 + runif(100, -200, 200)
ts <- ts(data)

plot(ts)
```




```r
print(length(ts))
```

```
[1] 100
```

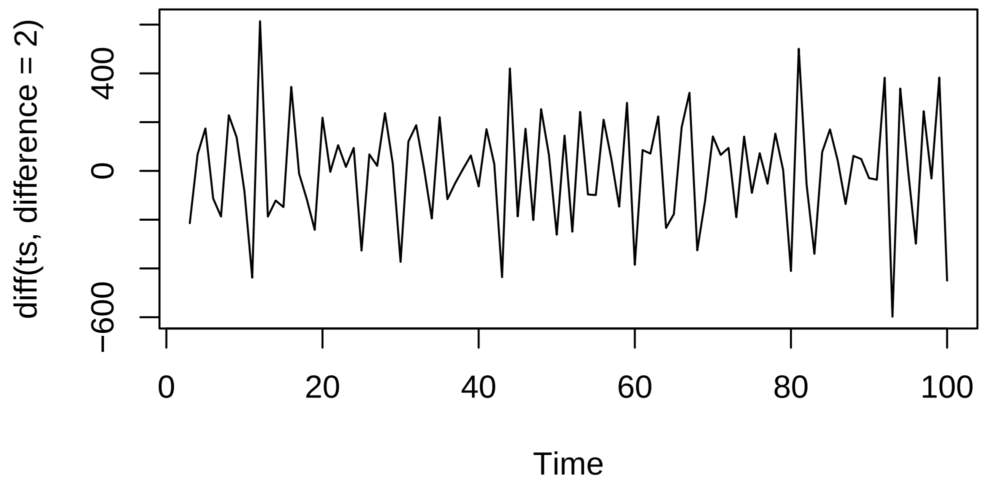
```r
plot(diff(ts))
```




```
print(length(diff(ts)))
```

```
[1] 99
```

```
plot(diff(ts, difference=2))
```



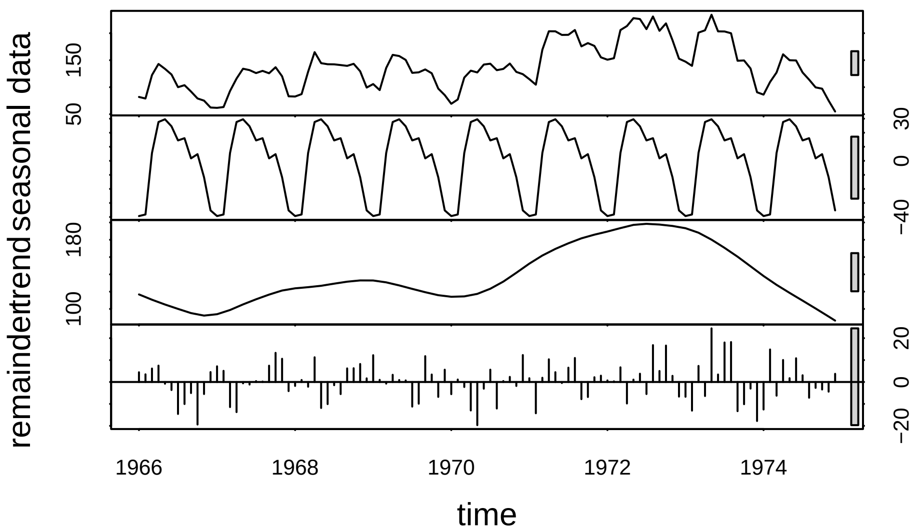

```
print(length(diff(ts, difference=2)))
```

```
[1] 98
```

We can remove the linear trend with two times backward differencing. The length of the ts increases by 2.

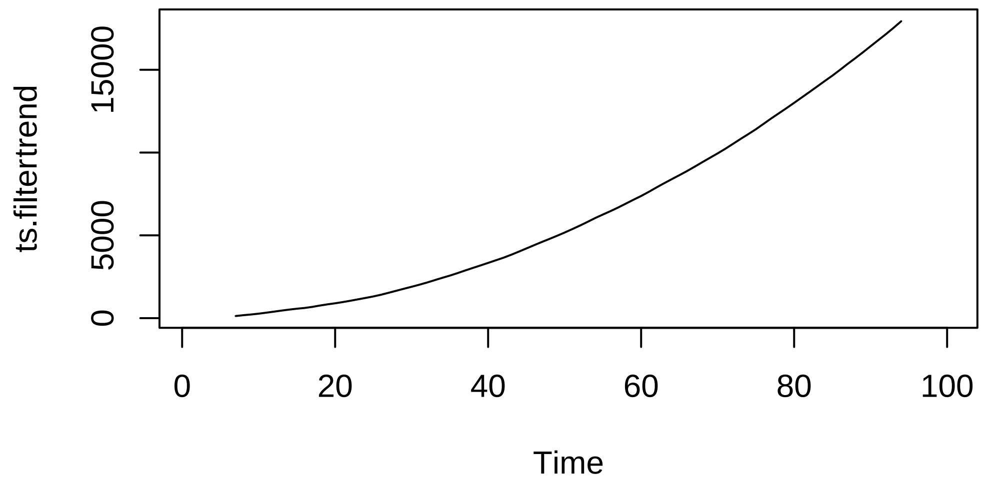
Exercise 2.2

```
hstart <- read.table("/home/nils/dev/mscids-notes/fs26/rtp/data/hstart.dat")

ts.hstart <- ts(
  data = hstart$V1,
  start = c(1966, 1), # January 1966
  end = c(1974, 12),  # December 1974
  frequency = 12      # Monthly data
)

decomp <- stl(ts.hstart, s.window = "periodic")
plot(decomp)
```



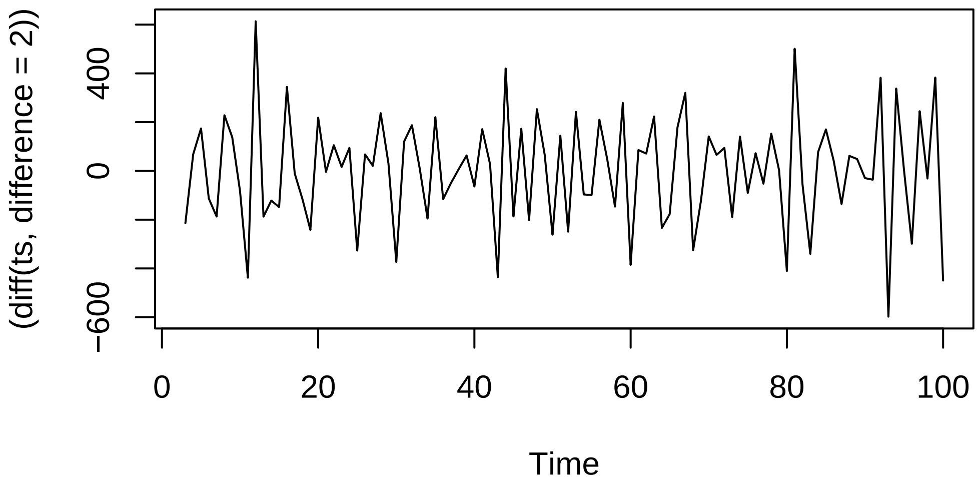
b

```r
weights <- c(0.5, rep(1, 11), 0.5) / 12
ts.filtertrend <- filter(
  ts,
  filter = weights,
  sides = 2
)

plot(ts.filtertrend)
```



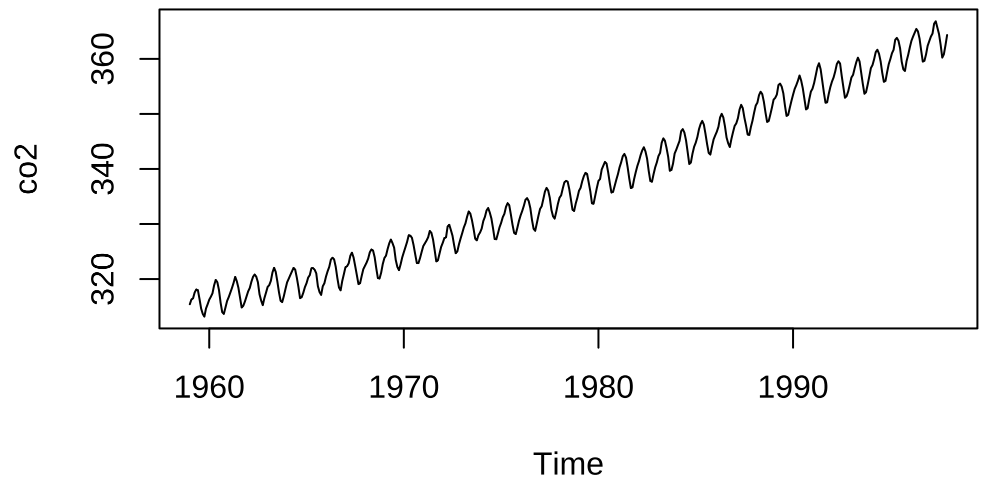
c

```r
hstart <- read.table("/home/nils/dev/mscids-notes/fs26/rtp/data/hstart.dat")

ts.hstart <- ts(
  data = hstart$V1,
  start = c(1966, 1), # January 1966
  end = c(1974, 12),  # December 1974
  frequency = 12      # Monthly data
)

plot((diff(ts, difference=2)))
```

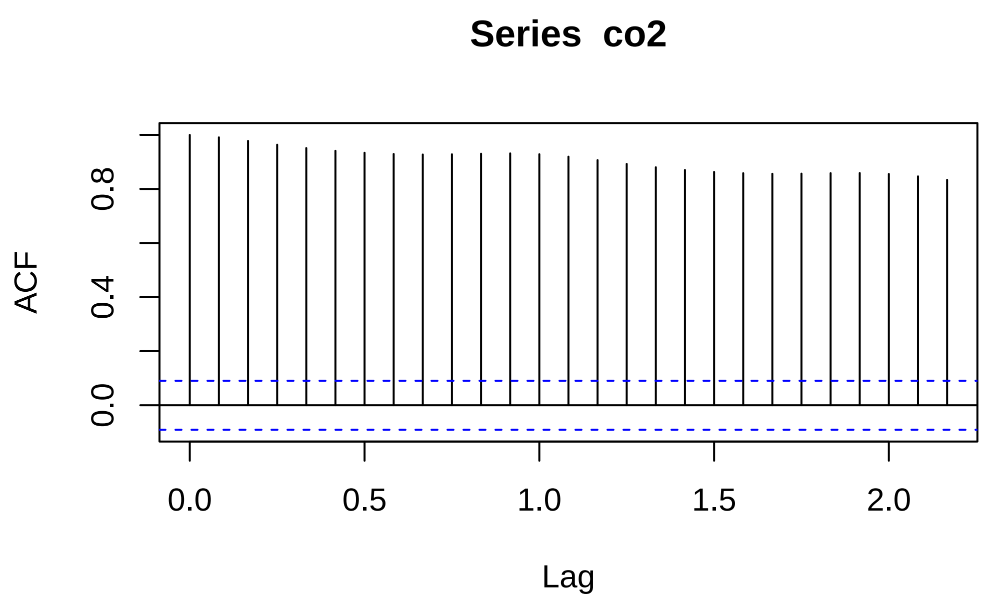


Exercise 2.3

```
data(co2)

plot(co2)
```


co2
320
340
360
1960
1970
1980
1990
Time

```
acf(co2)
```


Series co2
ACF
0.0
0.4
0.8
0.0
0.5
1.0
1.5
2.0
Lag

Series 3

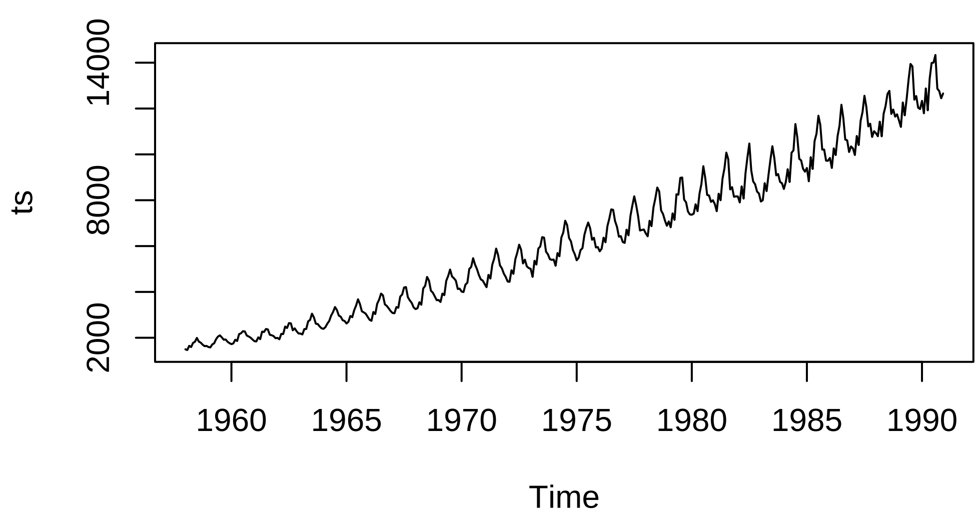
Exercise 3.1

A -> 1 B -> 3 C -> 2 D -> 4

Exercise 3.2

a

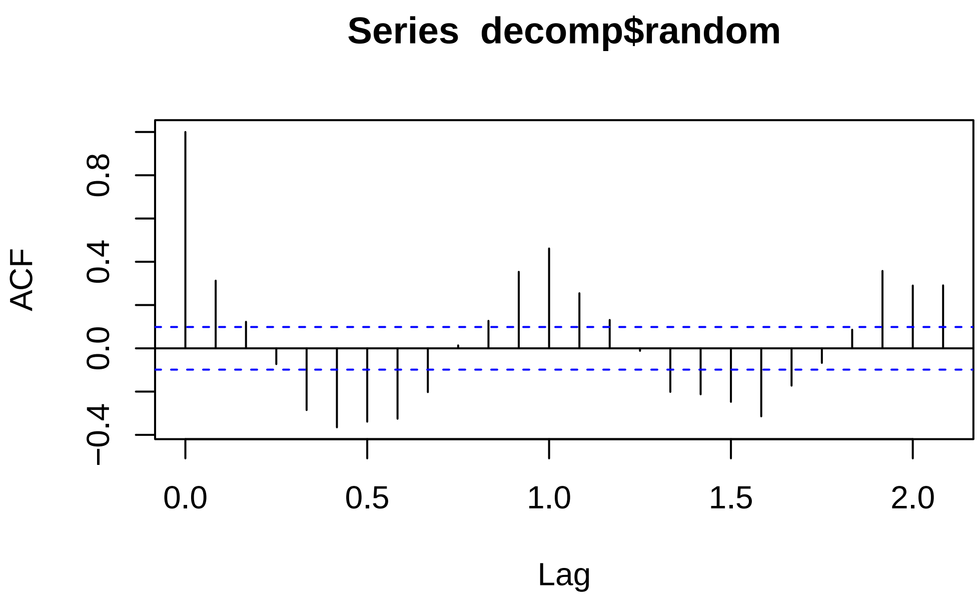
```
cbe <- read.table("/home/nils/dev/mscids-notes/fs26/rtp/data/cbe.dat", header=TRUE)

ts <- ts(
  data=cbe$elec,
  start=c(1958, 1),
  end=c(1990, 12),
  frequency=12
)

plot(ts)
```



It is not meaningful to interpret the correlogram of this time series because the process is not stationary.

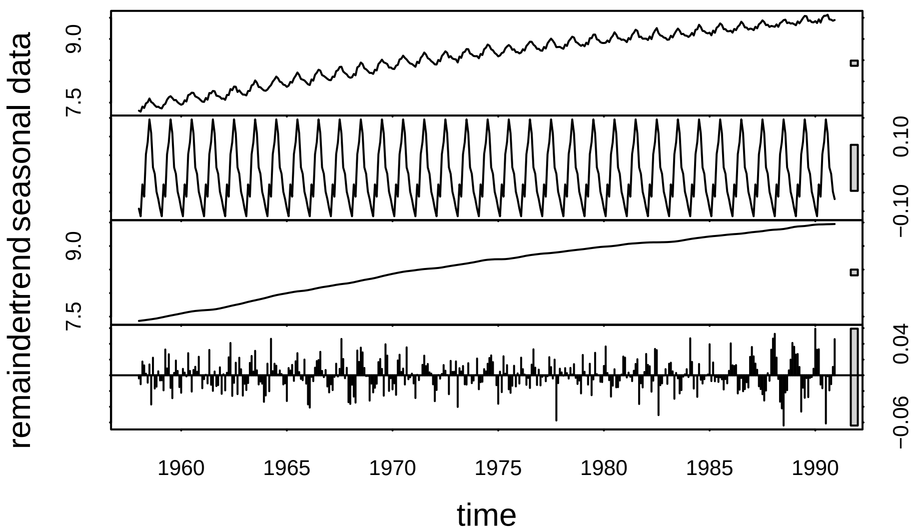
b

```
decomp <- decompose(ts, type = "multiplicative")
acf(decomp$random, na.action = na.pass)
```

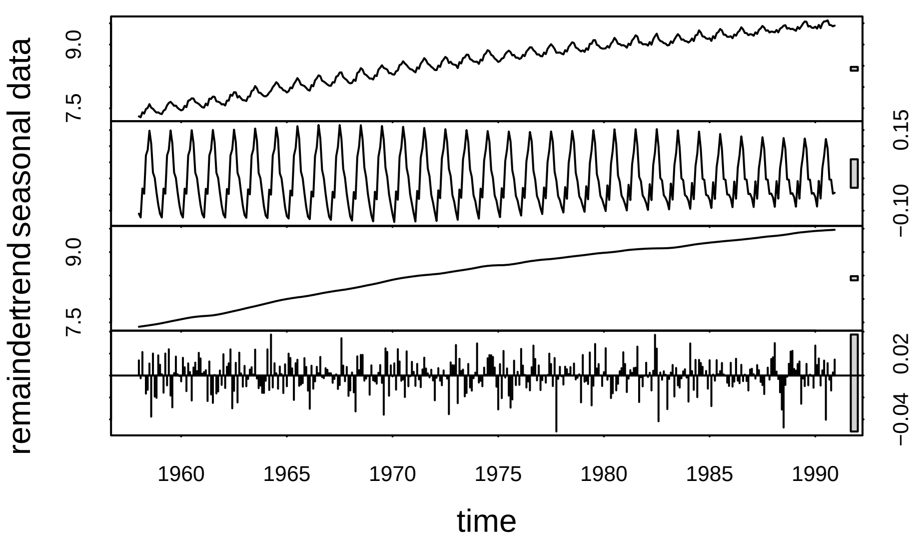


c

```
elec.stl1 <- stl(log(ts), s.window="periodic")
plot(elec.stl1)
```




```
elec.stl2 <- stl(log(ts), s.window=13)
plot(elec.stl2)
```



By simply log-transforming the data does make the process stationary.

Series 4

Exercise 4.1

a

```
data <- arima.sim(list(ar = c(0.8)), n = 1000)
```

b

```
ARMAacf(ar=c(0.8), lag.max = 10)
```

```
        0         1         2         3         4         5         6         7 
1.0000000 0.8000000 0.6400000 0.5120000 0.4096000 0.3276800 0.2621440 0.2097152 
        8         9        10 
0.1677722 0.1342177 0.1073742 
```

c

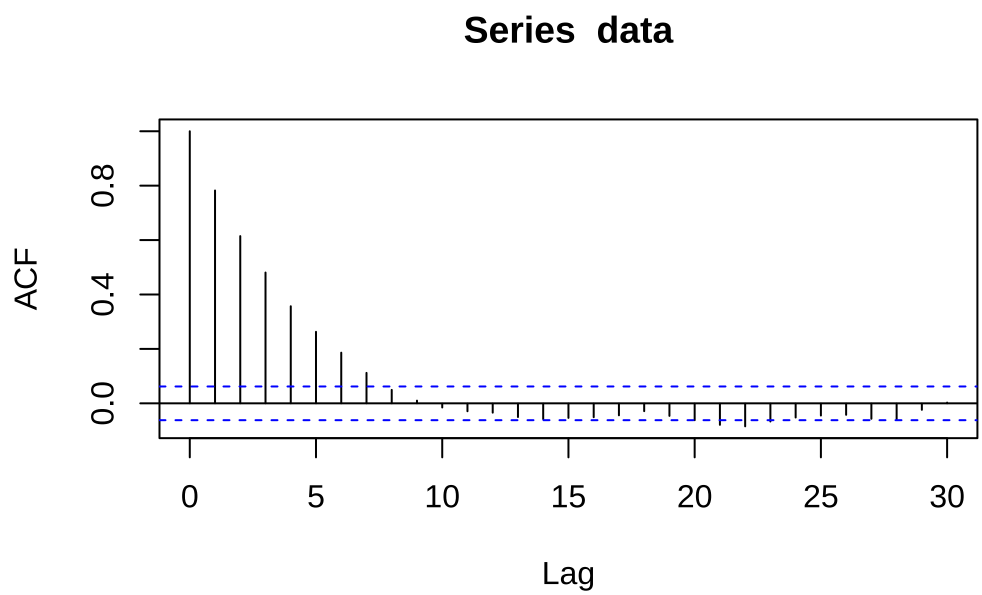
$p(k) = a_1^k = 0.8^k$

d

```
ARMAacf(ar=c(0.8), lag.max = 10)
```

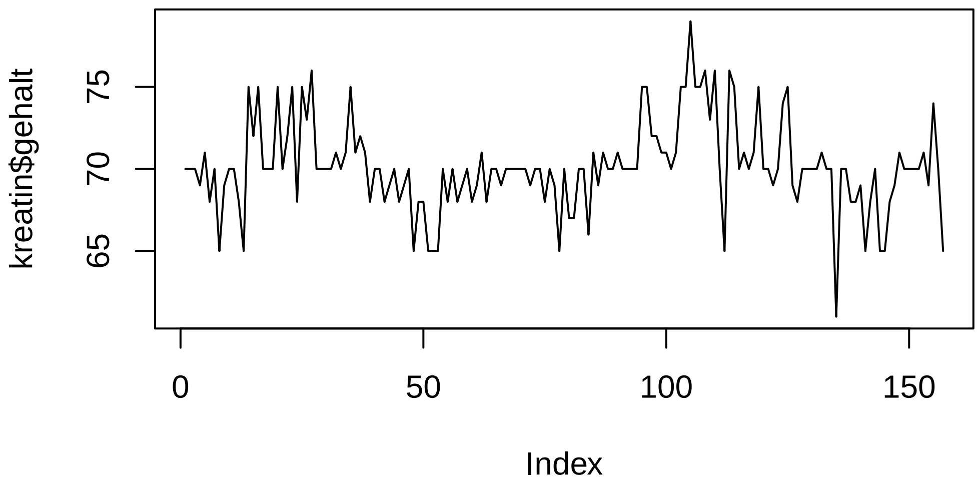
```
        0         1         2         3         4         5         6         7 
1.0000000 0.8000000 0.6400000 0.5120000 0.4096000 0.3276800 0.2621440 0.2097152 
        8         9        10 
0.1677722 0.1342177 0.1073742 
```

```
acf(data)
```



Exercise 4.2

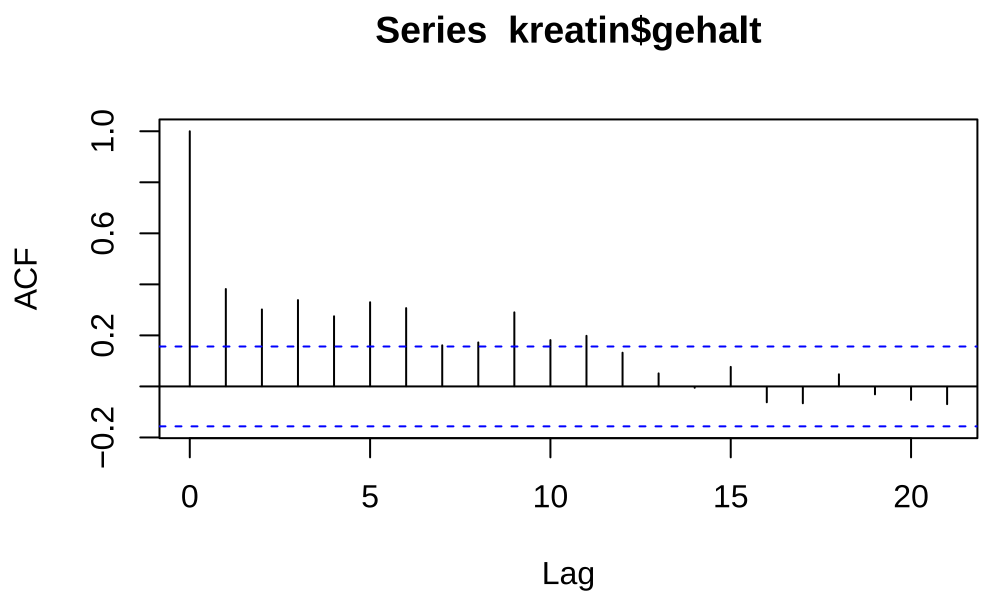
```
kreatin <- read.table("/home/nils/dev/mscids-notes/fs26/rtp/data/kreatin.dat", header=TRUE)

plot(kreatin$gehalt, type="l")
```



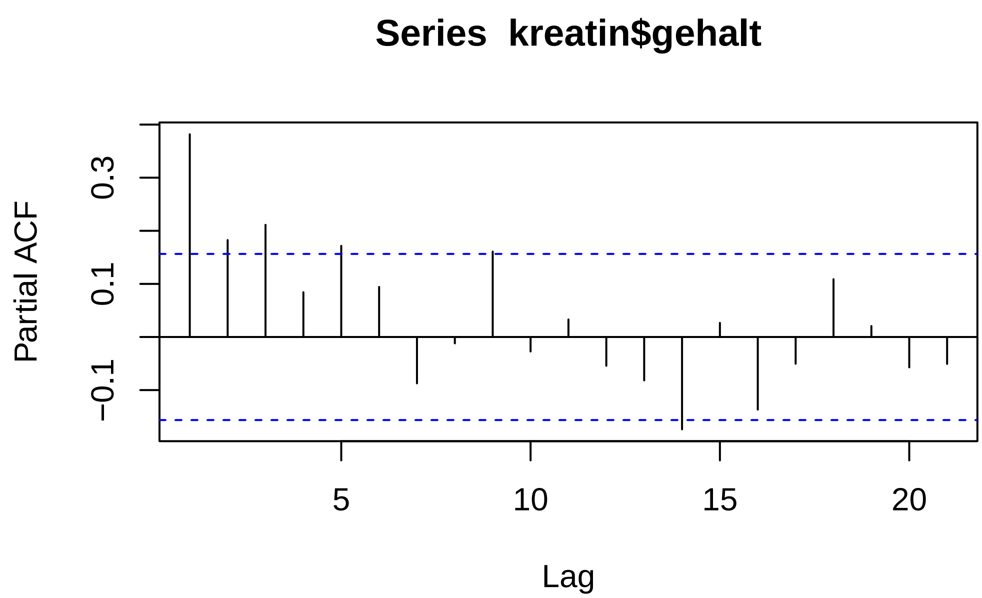
The process should follow white noise.

b

```r
acf(kreatin$gehalt)
```




```r
pacf(kreatin$gehalt)
```



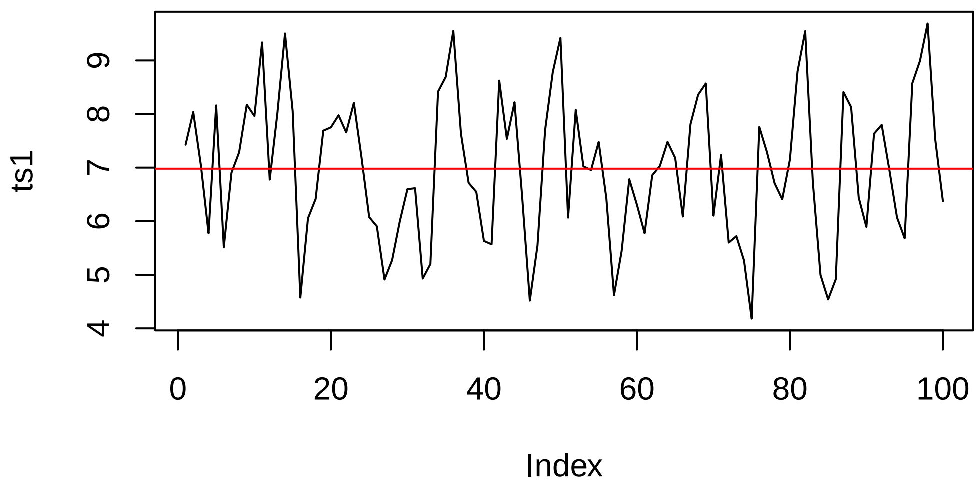
White noise fits.

Exercise 4.3

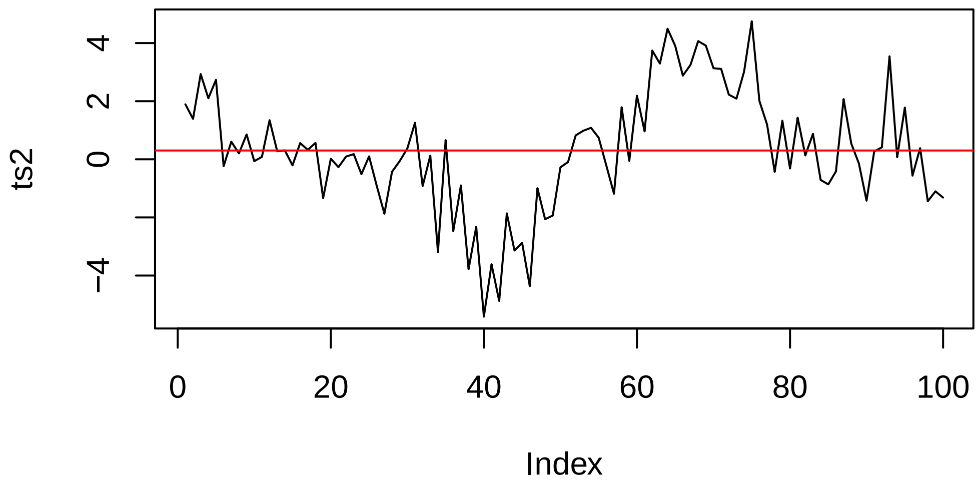
```
data <- read.table("/home/nils/dev/mscids-notes/fs26/rtp/data/ts_S3_A2.dat", header=TRUE)

ts1 <- data$ts1
ts2 <- data$ts2
```

a

```
plot(ts1, type="l")
abline(h=mean(ts1), col="red")
```




```
plot(ts2, type="l")
abline(h=mean(ts2), col="red")
```



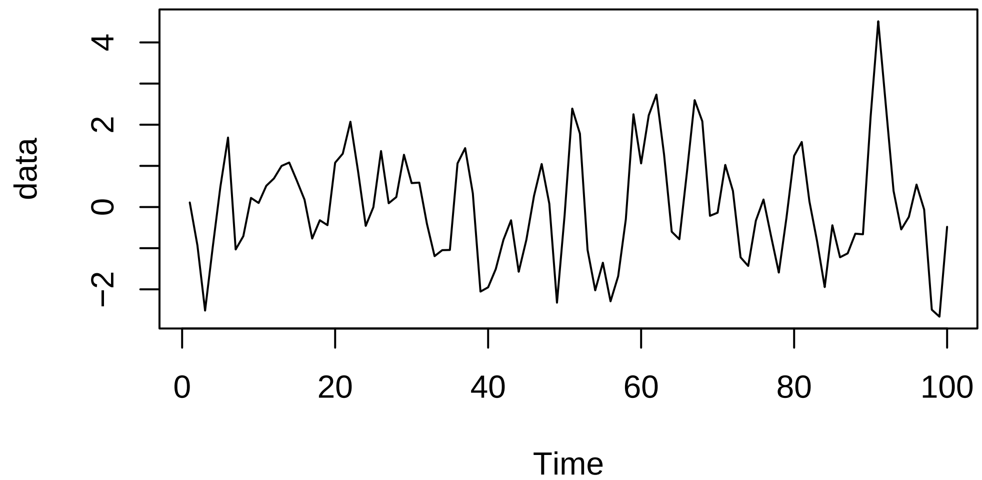
$ts1$ seems stationary, $ts2$ not.

Series 5

Exercise 5.1

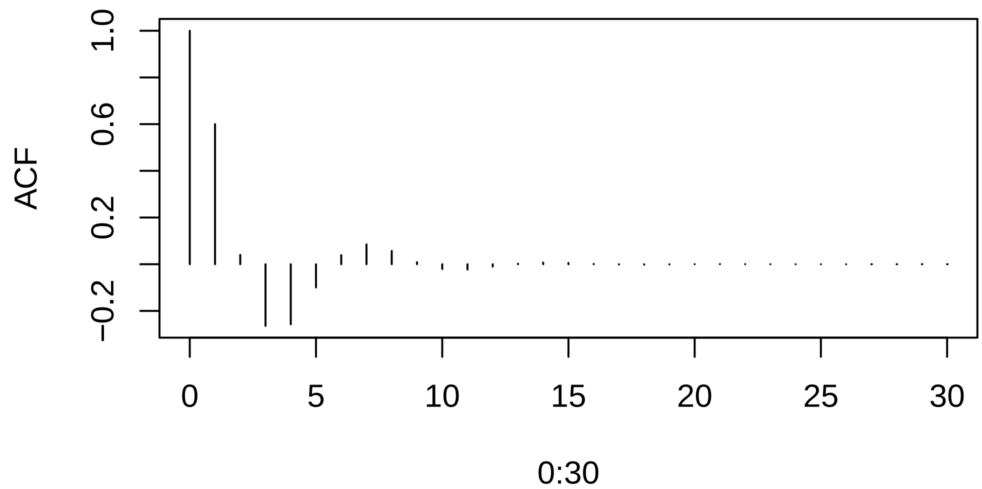
a

```
data <- arima.sim(list(ar = c(0.9, -0.5)), n = 100)
plot(data)
```

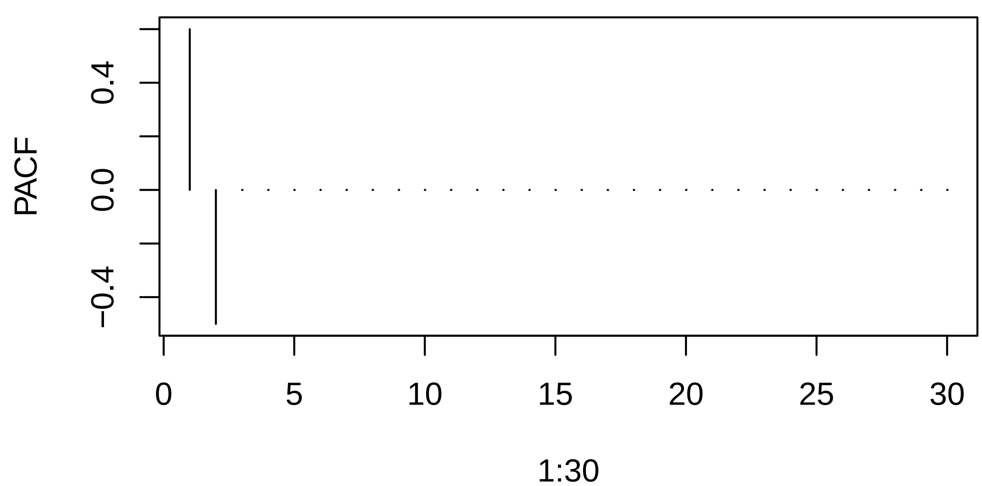


II

```
## Theoretical autocorrelations for X_t = 0.9
## X_t-1 - 0.5 X_t-2 + e_t
plot(0:30, ARMAacf(ar = c(0.9, -0.5), lag.max = 30), type = "h", ylab = "ACF")
```

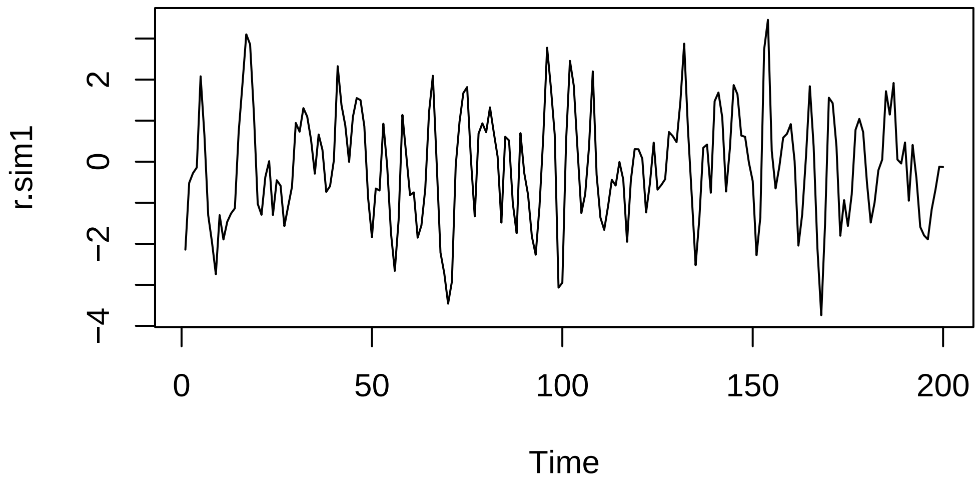



```r
## Theoretical partial autocorrelations
plot(1:30, ARMAacf(ar = c(0.9, -0.5), lag.max = 30, pacf = TRUE), type = "h", ylab = "PACF")
```



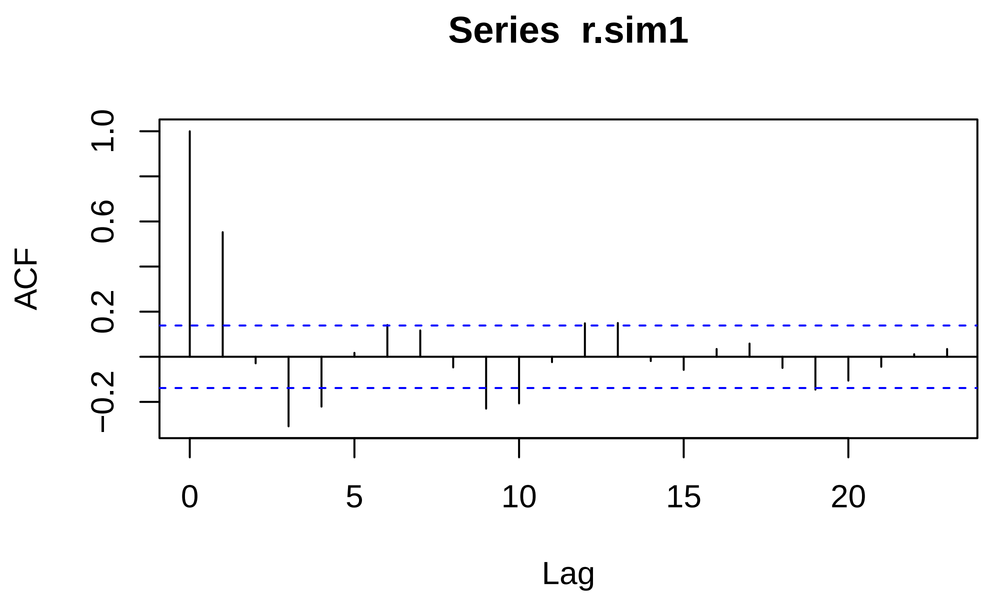
III

```r
r.sim1 <- arima.sim(n = 200, model = list(ar = c(0.9, -0.5)))
plot(r.sim1)
```

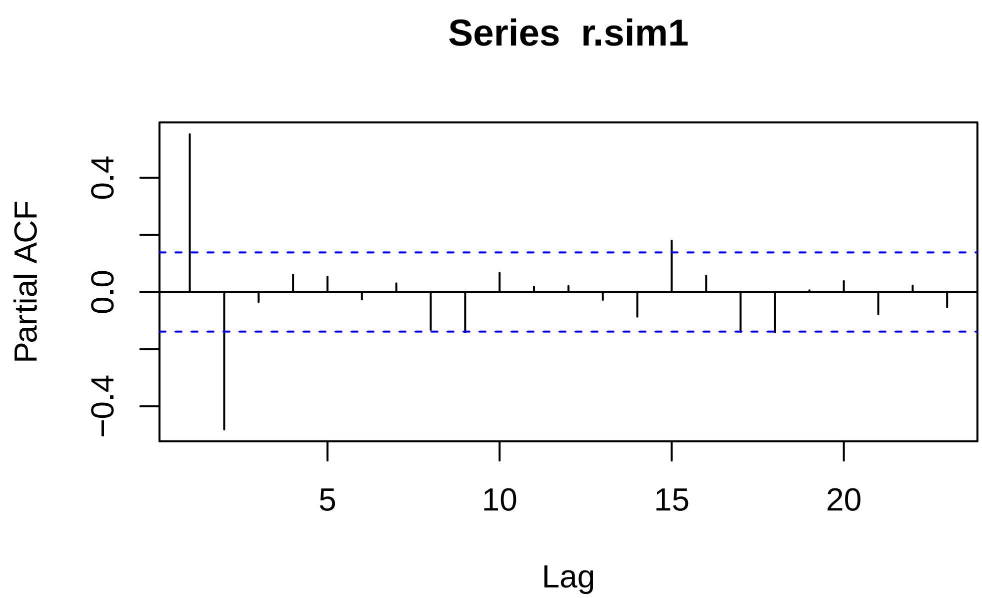


IV

```r
acf(r.sim1)
```


Series r.sim1
ACF
1.0
0.6
0.2
−0.2
0
5
10
15
20
Lag

```
pacf(r.sim1)
```


Series r.sim1
Partial ACF
0.4
0.0
−0.4
5
10
15
20
Lag

Exercise 5.2

a

I

```r
polyroot(c(1, 0.5, 2))
```

```
[1] -0.125+0.6959705i -0.125-0.6959705i
```

The process is not stationary

II

The process is not stationary since $a_1 = 1$.

b

```r
polyroot(c(1, 0.5, 0.26))
```

```
[1] -0.9615385+1.709268i -0.9615385-1.709268i
```

a_2 has to be < 0.25.

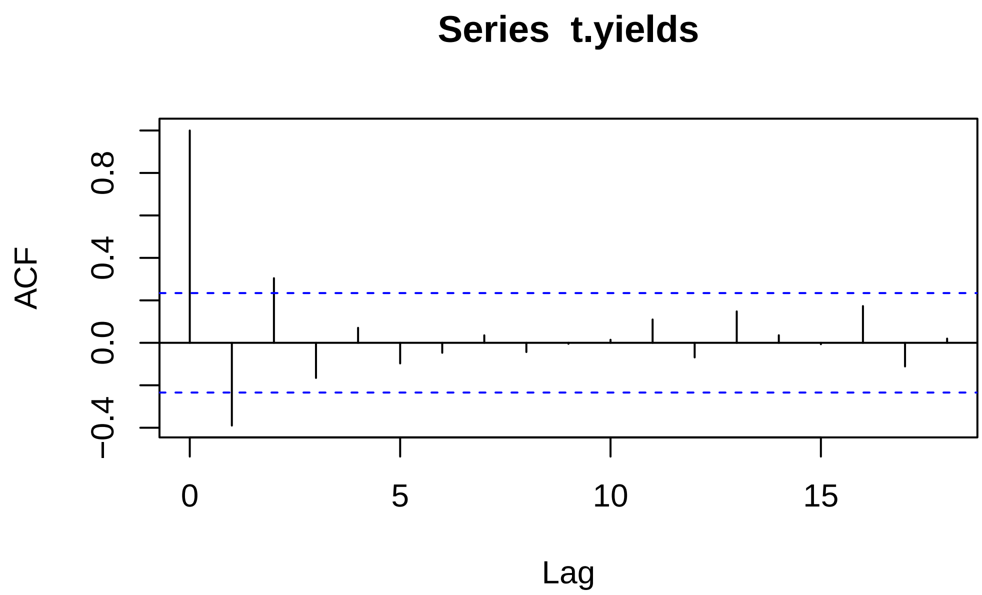
Series 6

Exercise 6.1

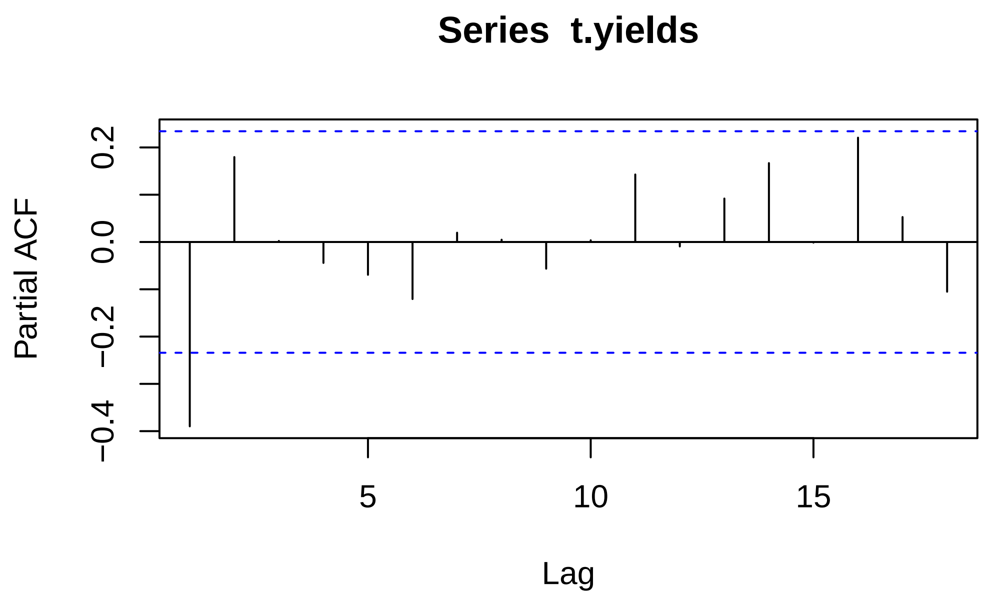
```r
yields <- read.table("/home/nils/dev/mscids-notes/fs26/rtp/data/yields.dat", header = FALSE)
t.yields <- ts(yields[, 1])
```

a

```r
acf(t.yields)
```




```r
pacf(t.yields)
```



The process can be modeled by a $AR(2)$.

b

```
r.yw <- ar(t.yields, method = "yw", order.max = 2)
r.yw$resid
```

```
Time Series:
Start = 1
End = 70
Frequency = 1
 [1]          NA          NA -23.2701676   8.5624227  -1.7185396   5.1017154
 [7]  10.3471780 -11.2034906   3.9364420   1.2089867  17.4563283  -9.2111662
[13]  -2.8577303  -6.3524133   2.2572167  -2.9311234  25.3567725  14.3859910
[19] -18.0787717  17.6571812   9.7250352   1.7201979   0.7723057   7.4553688
[25]  -3.0885436   2.3171776   1.8505361  -7.5446312 -27.8857718   0.6164494
[31]   6.0842678  18.0959587  11.4294237  20.8583891   5.3106389   2.4003827
[37]  -3.7281741  -0.3234123 -13.1089096  -2.4829289   0.5084360   2.3548536
[43]  -4.3282096   3.2157029   1.8505361   9.4553688  -3.4489132   8.6379522
[49]  -2.7310524  -4.0412731 -11.3891299   3.5161116   8.7481019 -10.3049653
[55]  -2.0635580   0.2900955 -18.1456261 -21.2240340   0.7913572  -2.3118589
```

```
[61]  14.3954080  -6.6314966  -8.3591694  10.8697651  -9.0885436   2.3982863
[67]  -6.4316031   0.8978066   6.7490614 -28.2653704
```

```r
str(r.yw)
```

```
List of 15
 $ order       : int 2
 $ ar          : num [1:2] -0.32 0.18
 $ var.pred    : num 120
 $ x.mean      : num 51.1
 $ aic         : Named num [1:3] 9.84 0.298 0
  ..- attr(*, "names")= chr [1:3] "0" "1" "2"
 $ n.used      : int 70
 $ n.obs       : int 70
 $ order.max   : num 2
 $ partialacf  : num [1:2, 1, 1] -0.39 0.18
 $ resid       : Time-Series [1:70] from 1 to 70: NA NA -23.27 8.56 -1.72 ...
 $ method      : chr "Yule-Walker"
 $ series      : chr "t.yields"
 $ frequency   : num 1
 $ call        : language ar(x = t.yields, order.max = 2, method = "yw")
 $ asy.var.coef: num [1:2, 1:2] 0.01444 0.00563 0.00563 0.01444
 - attr(*, "class")= chr "ar"
```

c

```r
r.burg <- ar(t.yields, method = "burg", order.max = 2)
r.burg$resid
```

```
Time Series:
Start = 1
End = 70
Frequency = 1
 [1]          NA          NA -23.0964972   8.1701783  -1.3785617   4.8728473
 [7]  10.5505765 -11.1978722   3.7977236   1.3403098  17.3914298  -8.9537665
[13]  -3.1238980  -6.2261379   2.0994588  -2.8089304  25.2454443  14.7685021
[19] -18.1260033  17.4687299  10.0560689   1.6430522   0.8457311   7.4219430
[25]  -2.9745155   2.2116763   1.9437802  -7.5780570 -27.9582418   0.3118307
[31]   6.2684408  18.0559266  11.6802172  20.8533189   5.5789133   2.3108038
```

```
[37]  -3.6390123  -0.4223073 -13.0529651  -2.6805103   0.5941087   2.3064707
[43]  -4.2697412   3.1267173   1.9437802   9.4219430  -3.3100186   8.5134115
[49]  -2.5474727  -4.1848533 -11.3514455   3.3500029   8.8893344 -10.2828313
[55]  -2.2022764   0.3468193 -18.1782727 -21.4299662   0.6474050  -2.2228832
[61]  14.3097254  -6.4014872  -8.5781282  10.8990986  -8.9745155   2.2181858
[67]  -6.2936738   0.7334424   6.8588212 -28.2490633
```

```r
str(r.burg)
```

```
List of 15
 $ order       : int 2
 $ ar          : num [1:2] -0.332 0.183
 $ var.pred    : num 113
 $ x.mean      : num 51.1
 $ aic         : Named num [1:3] 11.039 0.385 0
  ..- attr(*, "names")= chr [1:3] "0" "1" "2"
 $ n.used      : int 70
 $ n.obs       : int 70
 $ order.max   : num 2
 $ partialacf  : num [1:2, 1, 1] -0.407 0.183
 $ resid       : Time-Series [1:70] from 1 to 70: NA NA -23.1 8.17 -1.38 ...
 $ method      : chr "Burg"
 $ series      : chr "t.yields"
 $ frequency   : num 1
 $ call        : language ar(x = t.yields, order.max = 2, method = "burg")
 $ asy.var.coef: num [1:2, 1:2] 0.0136 0.0053 0.0053 0.0136
 - attr(*, "class")= chr "ar"
```

d

```r
r.mle <- ar(t.yields, method = "mle", order.max = 2)
r.mle$resid
```

```
Time Series:
Start = 1
End = 70
Frequency = 1
 [1]            NA            NA -23.083437837   7.762346509  -1.201980697
 [6]   4.561565507  10.602738000 -11.334850093   3.581349256   1.337116357
[11]  17.216926017  -8.885802464  -3.460572312  -6.220191632   1.865930909
```

```
[16]  -2.816243379  25.041428463  14.930022474 -18.332496024  17.218498832
[21]  10.197216932   1.428770775   0.782162438   7.273056970  -3.008346873
[26]   2.007435328   1.906279662  -7.726943030 -28.119035966   0.003476264
[31]   6.334762287  17.898351229  11.739491795  20.694392418   5.637512389
[36]   2.088058169  -3.689772086  -0.617858931 -13.115858157  -2.934860904
[41]   0.570317679   2.153690370  -4.337204163   2.944199681   1.906279662
[46]   9.273056970  -3.326921661   8.292016980  -2.521379225  -4.423401367
[51] -11.422577628   3.121270050   8.899174969 -10.398085740  -2.418650744
[56]   0.292840720 -18.318244153 -21.679521763   0.471483903  -2.242329450
[61]  14.131552552  -6.345881671  -8.876375881  10.832761513  -9.008346873
[66]   1.963159691  -6.288177902   0.491225167   6.843044015 -28.364092605
```

```r
str(r.mle)
```

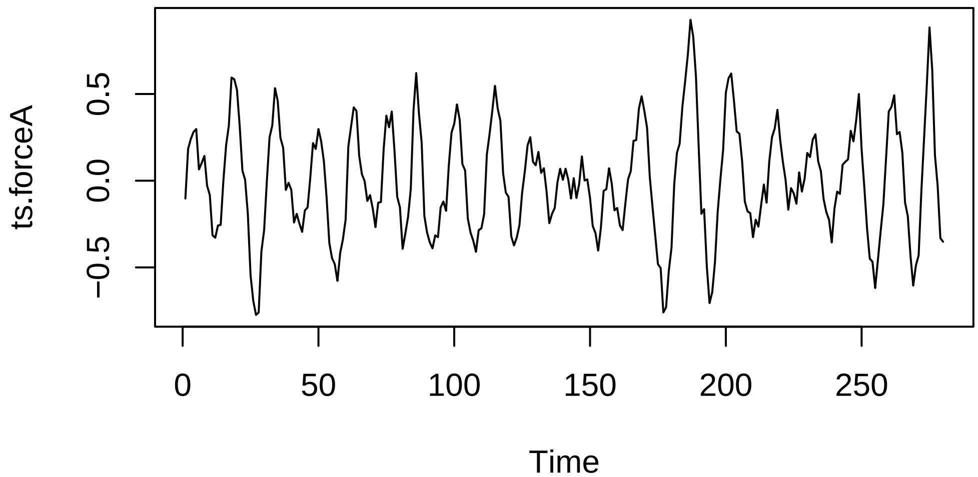
```
List of 15
 $ order       : int 2
 $ ar          : num [1:2] -0.341 0.187
 $ var.pred    : num 113
 $ x.mean      : num 51.2
 $ aic         : Named num [1:3] 10.808 0.301 0
  ..- attr(*, "names")= chr [1:3] "0" "1" "2"
 $ n.used      : int 70
 $ n.obs       : int 70
 $ order.max   : num 2
 $ partialacf  : NULL
 $ resid       : Time-Series [1:70] from 1 to 70: NA NA -23.08 7.76 -1.2 ...
 $ method      : chr "MLE"
 $ series      : chr "t.yields"
 $ frequency   : num 1
 $ call        : language ar(x = t.yields, order.max = 2, method = "mle")
 $ asy.var.coef: num [1:2, 1:2] 0.0136 0.0053 0.0053 0.0136
 - attr(*, "class")= chr "ar"
```

Exercise 6.2

```r
d.force <- read.table("/home/nils/dev/mscids-notes/fs26/rtp/data/kraft.dat", header = FALSE)
ts.force <- ts(d.force[, 1])
```

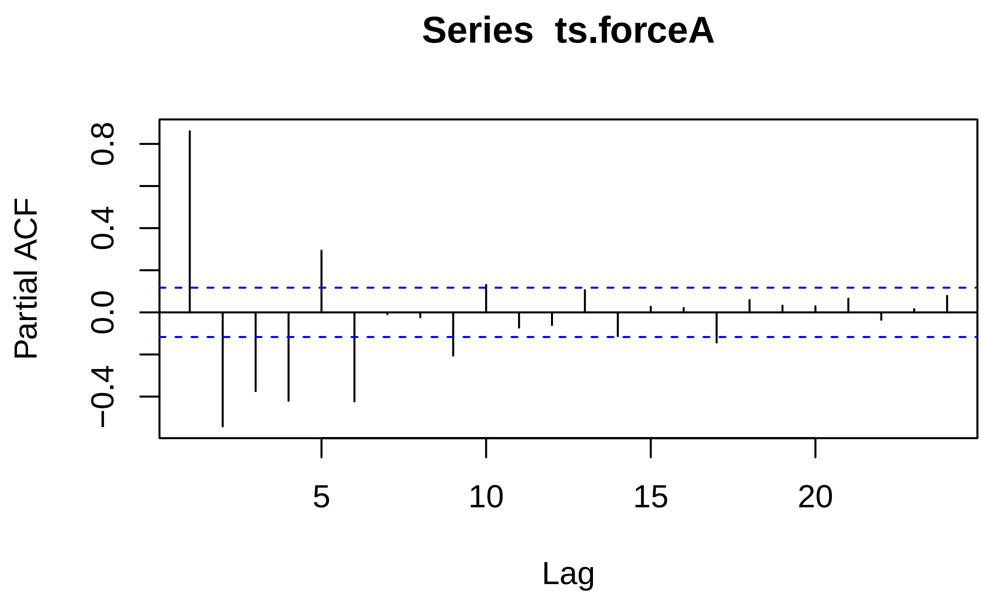
a

```r
ts.forceA <- window(ts.force, end = 280)
plot(ts.forceA)
```



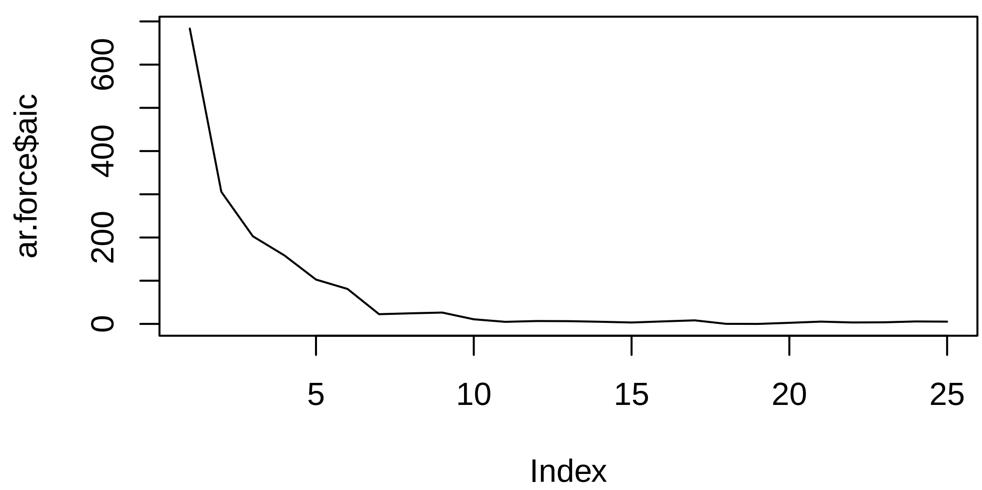
b

```r
pacf(ts.forceA)
```



We should use a $AR(7)$ model.

```
ar.force <- ar(ts.forceA, method = "ols")
plot(ar.force$aic, type="l")
```



c

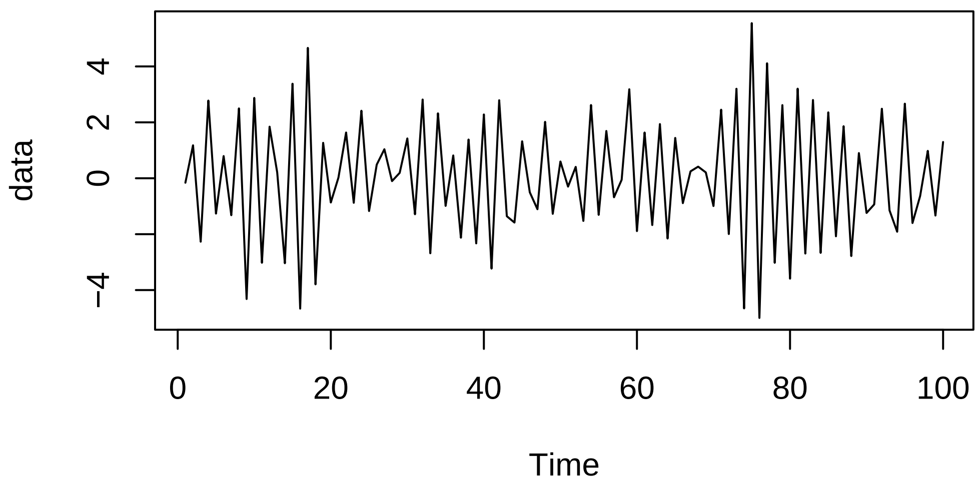
```r
ar.force <- arima(ts.forceA, order = c(7, 0, 0), method = "ML")
```

Series 7

Exercise 7.1

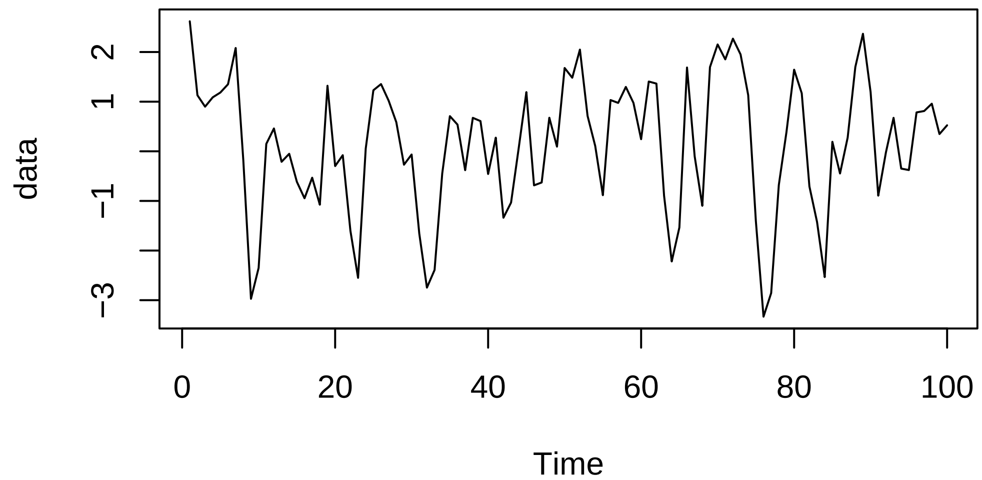
a

```r
data <- arima.sim(n = 100, model = list(ar = c(-0.75), ma = c(-0.3, 0.25)))
plot(data, type="l")
```



b

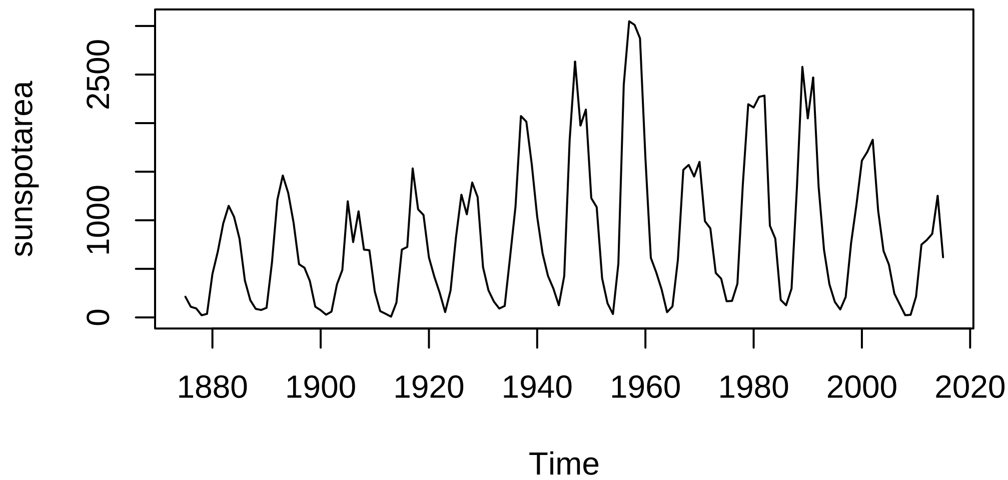
```r
data <- arima.sim(n = 100, model = list(ar = c(0.5, -0.3), ma = c(0.25)))
plot(data, type="l")
```



Exercise 7.2

a

```
library(fpp2)
data(sunspotarea)

plot(sunspotarea, type="l")
```



We should log-transform because the data are not stationary.

b

```
subset <- log(sunspotarea[1:100])
ar.sun<- ar(subset, method = "ols")
ar.sun
```

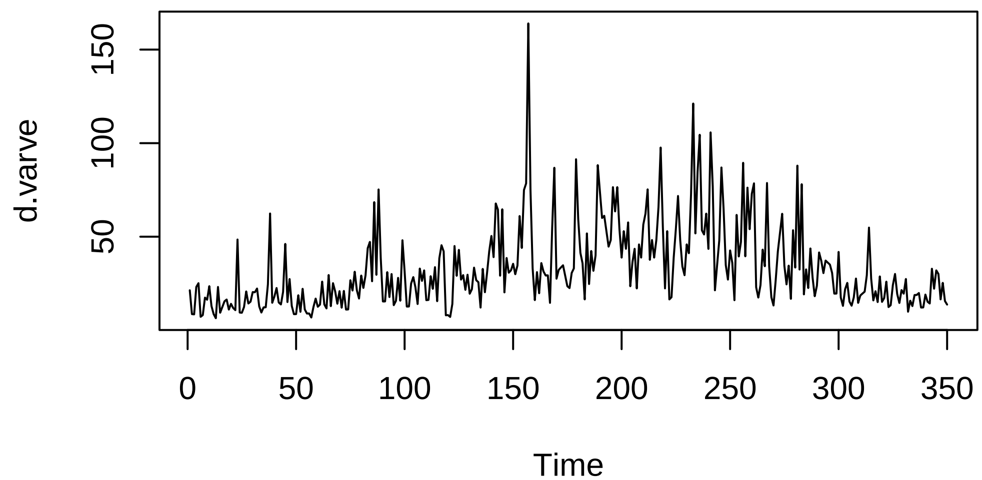
```
Call:
ar(x = subset, method = "ols")

Coefficients:
      1        2        3        4        5        6        7        8
 0.6986  -0.1578  -0.0555  -0.0398   0.0387  -0.1249   0.1495  -0.2127
      9       10
 0.1133   0.3531

Intercept: 0.0581 (0.06088)

Order selected 10  sigma^2 estimated as  0.3287
```

Exercise 7.3

```
t.url <- "http://stat.ethz.ch/Teaching/Datasets/WBL/varve.dat"
d.varve <- ts(scan(t.url)[201:550])
```

a

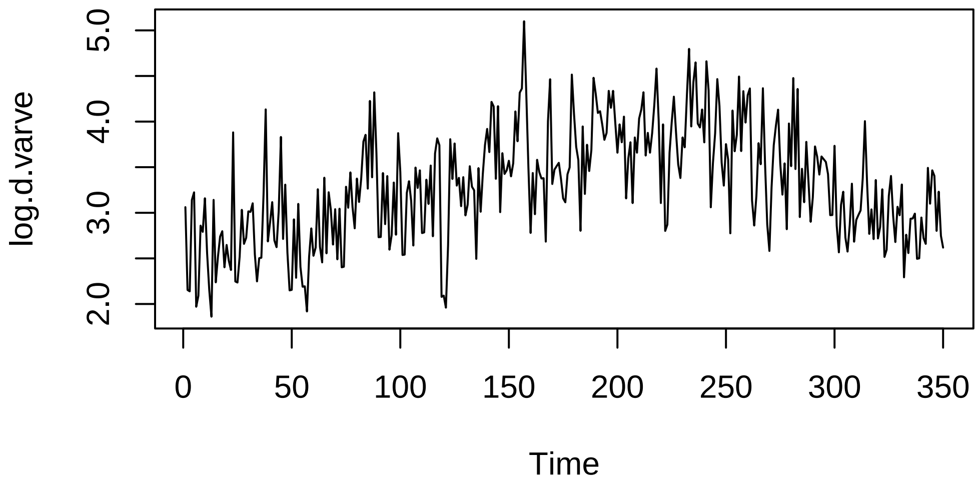
```
plot(d.varve, type="l")
```



We should log-transform the data because the process is not stationary.

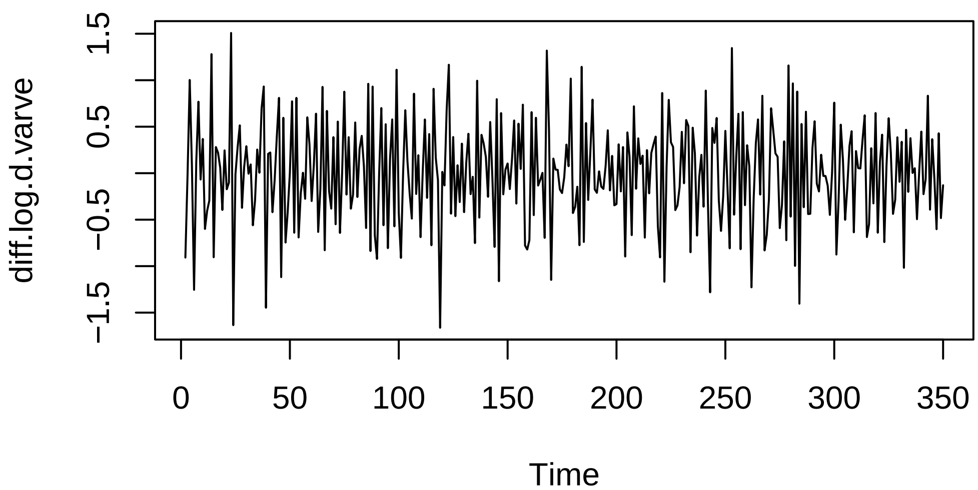
b

```
log.d.varve <- log(d.varve)
plot(log.d.varve, type="l")
```



Still not stationary.

```r
diff.log.d.varve <- diff(log.d.varve)
plot(diff.log.d.varve, type="l")
```



By applying differencing, we can make the process stationary.

Series 8

Exercise 8.1

a

```
arima.sim(model = list(order=c(2, 1, 2), ar=c(0.5, 0.5), ma=c(-0.4, 0.3)), n=100)
```

´Error in arima.sim(model = list(order = c(2, 1, 2), ar = c(0.5, 0.5),: 'ar' part of model is not stationary´

b

1. **Substitute the given coefficients**:

$$(1 - 0.5B - 0.5B^2)(1 - B)X_t = (1 - 0.4B + 0.3B^2)E_t$$

2. **Expand the left-hand side (AR polynomial multiplied by the difference operator)**:

$$(1 - B - 0.5B + 0.5B^2 - 0.5B^2 + 0.5B^3)X_t = (1 - 0.4B + 0.3B^2)E_t$$

3. **Simplify the expression**:

$$(1 - 1.5B + 0.5B^3)X_t = (1 - 0.4B + 0.3B^2)E_t$$

The resulting expression represents an $\textbf{ARMA}(3, 2)$ process with the following parameters:

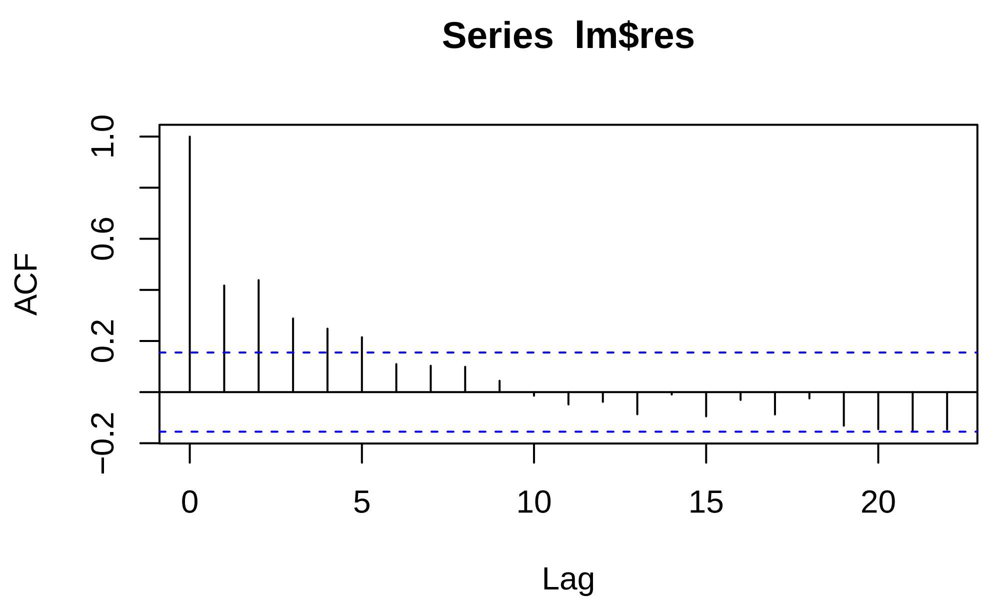
- **Autoregressive (AR) Coefficients:**
  - $\alpha_1^* = 1.5$
  - $\alpha_2^* = 0$
  - $\alpha_3^* = -0.5$
- **Moving Average (MA) Coefficients:**
  - $\beta_1 = -0.4$
  - $\beta_2 = 0.3$

Exercise 8.1

```
d.beluga <- read.table("http://stat.ethz.ch/Teaching/Datasets/WBL/beluga.dat", header = TRUE)
d.beluga <- ts(d.beluga)
```

a

```
lm <- lm(NURSING ~ PERIOD + BOUTS + LOCKONS + DAYNIGHT, data=d.beluga)
acf(lm$res)
```



looks more or less good.

b

```
r.burg <- ar(d.beluga, method = "burg")
str(r.burg)
```

```
List of 14
 $ order     : int 6
 $ ar        : num [1:6, 1:5, 1:5] 2.5182 -1.6528 -0.0945 -0.095 0.4912 ...
  ..- attr(*, "dimnames")=List of 3
  .. ..$ : chr [1:6] "1" "2" "3" "4" ...
  .. ..$ : chr [1:5] "PERIOD" "BOUTS" "NURSING" "LOCKONS" ...
  .. ..$ : chr [1:5] "PERIOD" "BOUTS" "NURSING" "LOCKONS" ...
 $ var.pred  : num [1:5, 1:5] 6.17e-04 -7.08e-04 -3.09e-03 -1.23e-03 8.31e-05 ...
  ..- attr(*, "dimnames")=List of 2
  .. ..$ : chr [1:5] "PERIOD" "BOUTS" "NURSING" "LOCKONS" ...
  .. ..$ : chr [1:5] "PERIOD" "BOUTS" "NURSING" "LOCKONS" ...
 $ x.mean    : Named num [1:5] 80.5 2.79 11.49 3.61 0.5
  ..- attr(*, "names")= chr [1:5] "PERIOD" "BOUTS" "NURSING" "LOCKONS" ...
 $ aic       : Named num [1:23] 3946 2628 242 157 130 ...
  ..- attr(*, "names")= chr [1:23] "0" "1" "2" "3" ...
 $ n.used    : int 160
 $ n.obs     : int 160
 $ order.max : num 22
 $ partialacf: num [1:22, 1:5, 1:5] 0.9997 -0.9993 0.762 0.3808 0.0846 ...
  ..- attr(*, "dimnames")=List of 3
  .. ..$ : chr [1:22] "1" "2" "3" "4" ...
  .. ..$ : chr [1:5] "PERIOD" "BOUTS" "NURSING" "LOCKONS" ...
  .. ..$ : chr [1:5] "PERIOD" "BOUTS" "NURSING" "LOCKONS" ...
 $ resid     : 'mts' num [1:160, 1:5] NA NA NA NA NA ...
  ..- attr(*, "tsp")= num [1:3] 1 160 1
  ..- attr(*, "dimnames")=List of 2
  .. ..$ : NULL
  .. ..$ : chr [1:5] "PERIOD" "BOUTS" "NURSING" "LOCKONS" ...
 $ method    : chr "Burg"
 $ series    : chr "d.beluga"
 $ frequency : num 1
 $ call      : language ar(x = d.beluga, method = "burg")
 - attr(*, "class")= chr "ar"
```

c

```
library(nlme) # Load the package containing the procedure gls()
r.bel.gls <- gls(NURSING ~ BOUTS + LOCKONS + DAYNIGHT + PERIOD, data = d.beluga, correlation

summary(r.bel.gls)
```

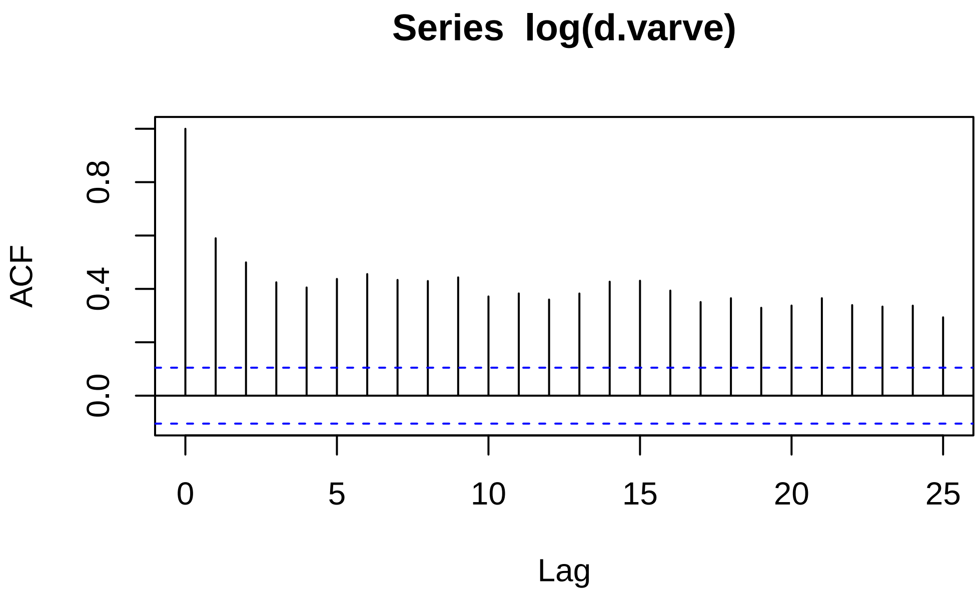
```
Generalized least squares fit by maximum likelihood
  Model: NURSING ~ BOUTS + LOCKONS + DAYNIGHT + PERIOD
  Data: d.beluga
       AIC      BIC    logLik
  565.0884 601.9905 -270.5442

Correlation Structure: ARMA(6,0)
 Formula: ~PERIOD
 Parameter estimate(s):
       Phi1        Phi2        Phi3        Phi4        Phi5        Phi6
 0.26391489  0.33076391 -0.01473528  0.08533177  0.11805117 -0.08347252

Coefficients:
                 Value Std.Error   t-value p-value
(Intercept)  1.5947050 0.8171076  1.951646  0.0528
BOUTS        0.1634638 0.3265945  0.500510  0.6174
LOCKONS      2.5699567 0.1888387 13.609269  0.0000
DAYNIGHT    -0.2836475 0.1727329 -1.642116  0.1026
PERIOD       0.0032607 0.0071800  0.454133  0.6504

 Correlation:
         (Intr) BOUTS  LOCKON DAYNIG
BOUTS    -0.271
LOCKONS  -0.082 -0.810
DAYNIGHT -0.038 -0.119  0.060
PERIOD   -0.652 -0.207  0.206  0.020

Standardized residuals:
        Min          Q1         Med          Q3         Max
-2.69029710 -0.55948989  0.02751544  0.62878372  2.48793230

Residual standard error: 1.595643
Degrees of freedom: 160 total; 155 residual
```

```r
d.resid <- ts(resid(r.bel.gls))
```

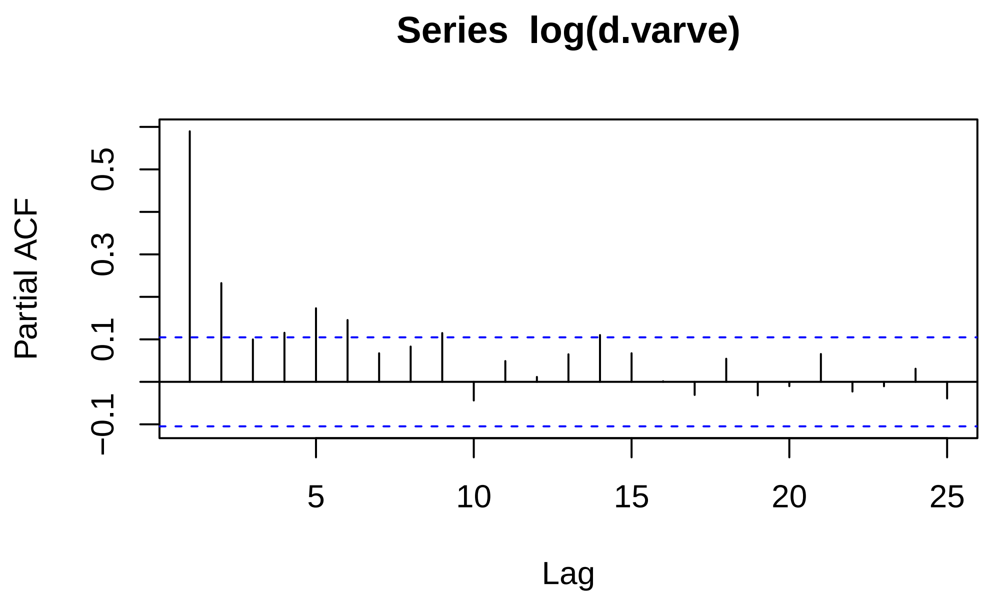
Exercise 8.3

a

```r
acf(log(d.varve))
```




```r
pacf(log(d.varve))
```




```r
r.varve <- arima(log(d.varve), order = c(0, 0, 9))
```

b

```r
r.varve
```

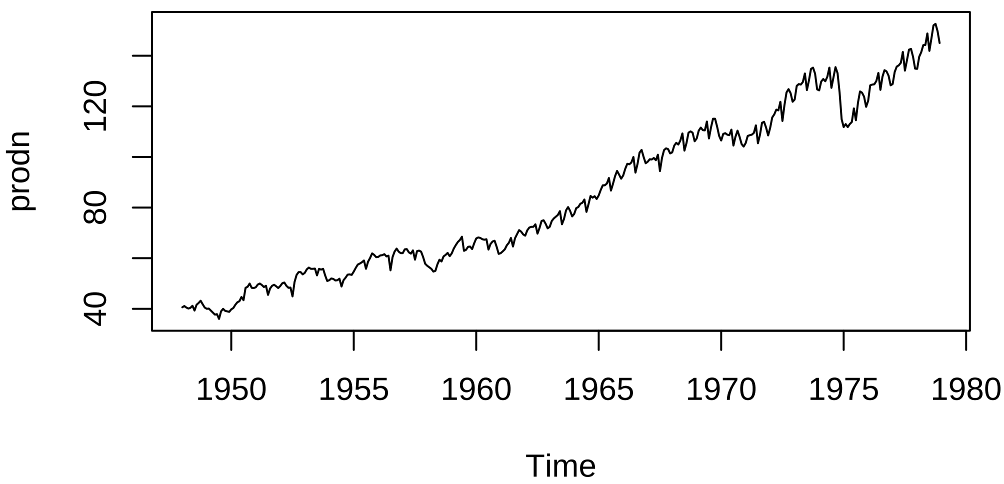
```

Call:
arima(x = log(d.varve), order = c(0, 0, 9))

Coefficients:
         ma1     ma2     ma3     ma4     ma5     ma6     ma7     ma8     ma9
      0.4074  0.3063  0.2006  0.1610  0.1934  0.2236  0.1694  0.1453  0.1674
s.e.  0.0536  0.0572  0.0594  0.0612  0.0587  0.0593  0.0561  0.0530  0.0522
      intercept
         3.2892
s.e.     0.0753

sigma^2 estimated as 0.228:  log likelihood = -238.25,  aic = 498.49
```

Series 9

Exercise 9.1

a

```
library(astsa)
data(prodn)

plot(prodn, type="l")
```

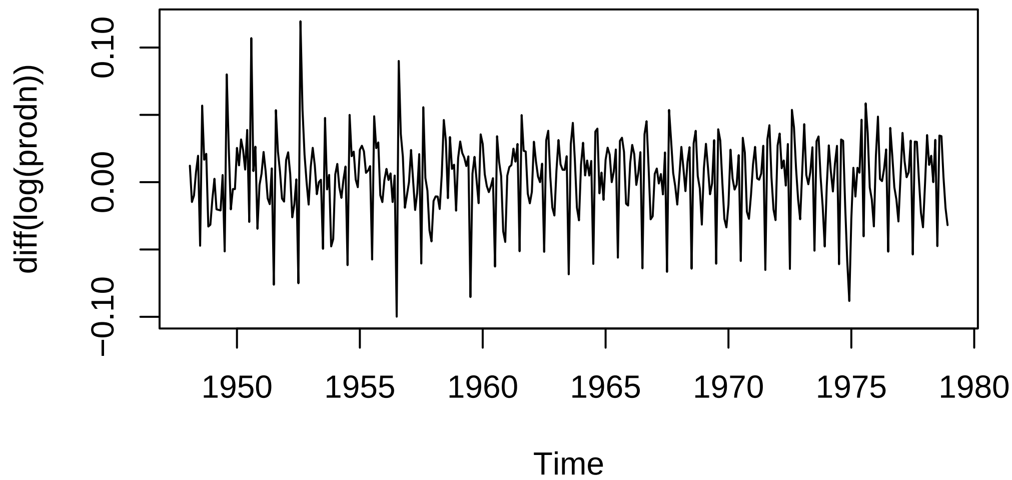


No constance variance. The process is not stationary.

b

We can apply tome transformations

```
plot(diff(log(prodn)))
```



c

```r
arima(prodn, order = c(1, 1, 1), seasonal = c(1, 1, 1))
```

```
Call:
arima(x = prodn, order = c(1, 1, 1), seasonal = c(1, 1, 1))

Coefficients:
         ar1      ma1     sar1     sma1
      0.5767  -0.2783  -0.0352  -0.6700
s.e.  0.1178   0.1376   0.0664   0.0424

sigma^2 estimated as 1.426:  log likelihood = -577.01,  aic = 1164.02
```

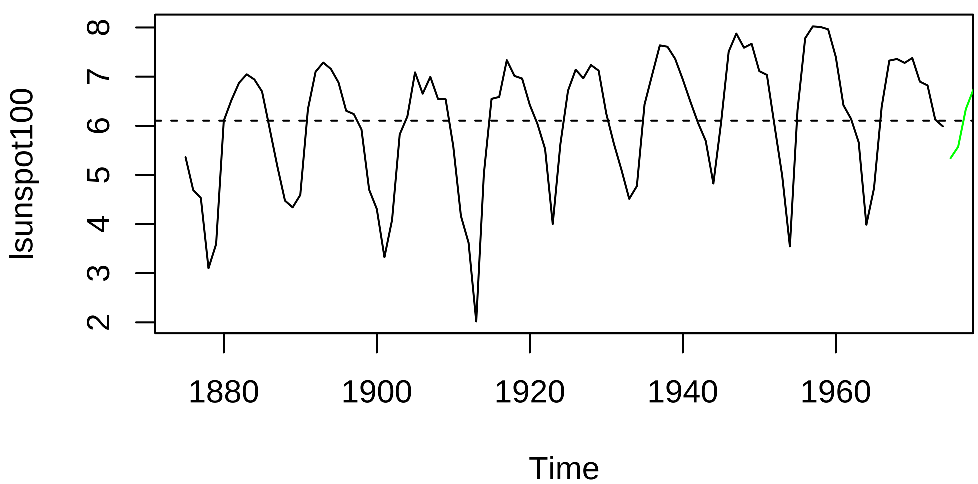
Series 10

Exercise 10.1

```
library(fpp2)
lsunspot100 <- window(log(sunspotarea), start = 1875, end = 1974)
fit.ar10 <- arima(lsunspot100, order = c(10, 0, 0))
```

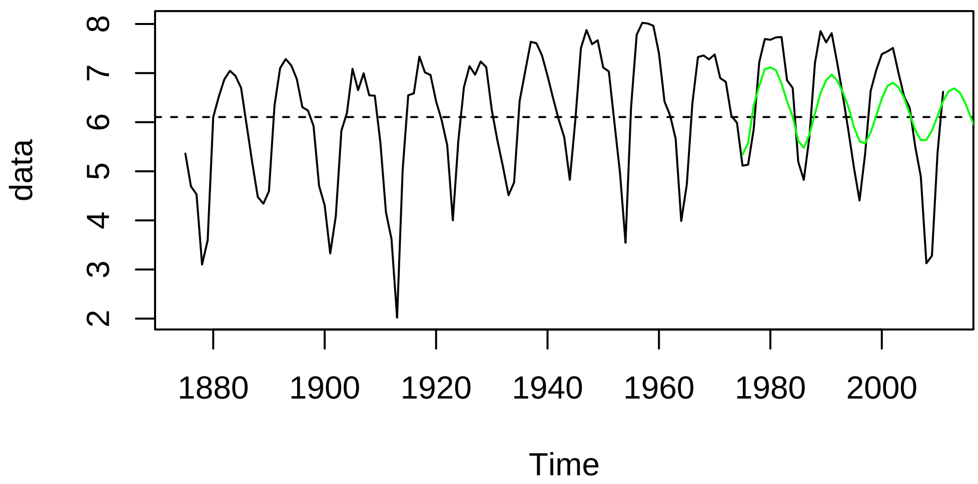
a

```
pred <- predict(fit.ar10, n.ahead = 100)
plot(lsunspot100)
abline(h = fit.ar10$coef["intercept"], lty = 2)
lines(pred$pred, col = "green")
```



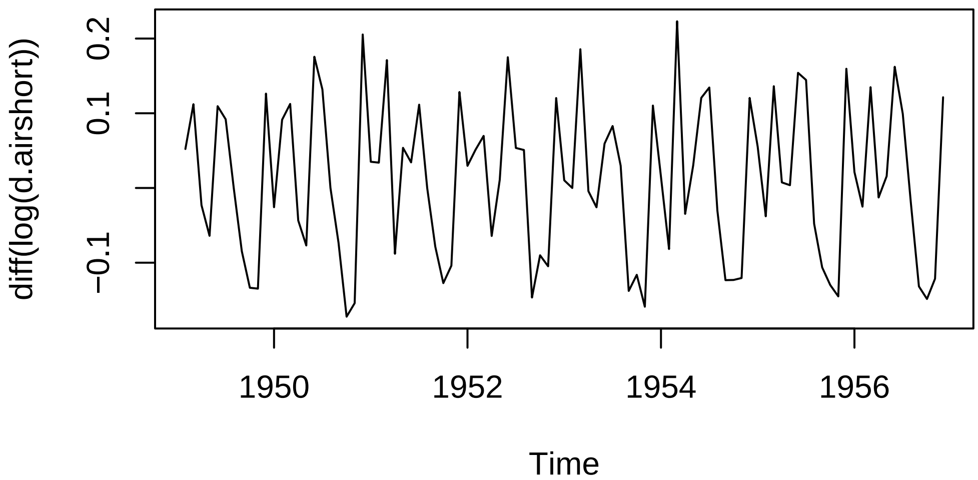
b

```r
data <- lsunspot100 <- window(log(sunspotarea), start = 1875, end = 2011)
plot(data)
abline(h = fit.ar10$coef["intercept"], lty = 2)
lines(pred$pred, col = "green")
```



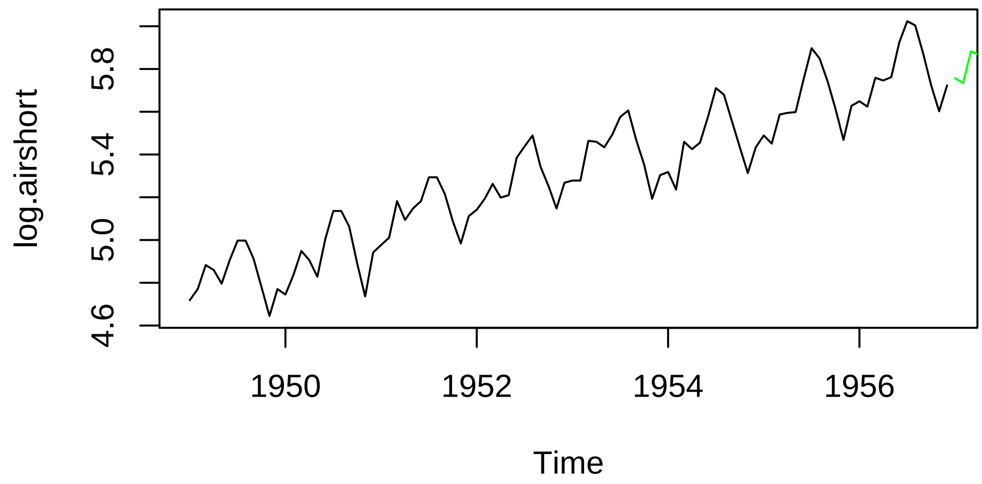
Exercise 10.2

```r
d.air <- AirPassengers
d.airshort <- window(d.air, end = c(1956, 12))
plot(diff(log(d.airshort)))
```



a

```r
log.airshort <- log(d.airshort)
fit.sarima <- arima(log.airshort,
                    order = c(0, 1, 1),
                    seasonal = list(
                      order = c(0, 1, 1),
                      period = 12
                      )
                    )
pred.sarima <- predict(fit.sarima, n.ahead = 48)

plot(log.airshort)
abline(h = fit.sarima$coef["intercept"], lty = 2)
lines(pred.sarima$pred, col = "green")
```



Series 11

Exercise 11.1

a

I

Not stationary

II

Statonary

III

Stationary